

A230 – Open Questions: The Residuals of the Series

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.1 Purpose: Residuals in One Place

[STIPULATION] Every document of the series closes with “Open Questions”. This document collects these residuals in one place, so that the open frontier of the theory is visible as a whole – not scattered, not glossed over.

A residual is not a weakness to be hidden. It is the edge at which work continues. A theory without named open points is either finished or dishonest; this one is neither.

Note: This edition reflects the state after the A-series extension 2026-07 (Blocks 0–3 complete, A095 and A192 new, several points raised or closed). Points previously listed as open but since closed are no longer carried here.

.2 Closed Points (Since Last Edition)

[K] The following points were open in earlier editions of A230 and have been closed in this session:

- $g_R = 0$ **from torus projection**. Was [S]. Now [B]: the Fourier projector P_+ annihilates the $|1, -1\rangle$ state algebraically; $g_R = \langle 0|H|0\rangle = 0$ follows from the kernel of P_+ (A095).
- **CHSH prefactor** $\xi/(2\pi)$. Was unexplained. Now [B]: 2π is the circumference of the unit circle; the phase correction per measurement is $\xi/(2\pi)$ (A160).
- **Cumulative CHSH formula** $\text{CHSH}^{\text{TO}}(n)$. Now [B]: $\xi \ln n/D_f$ integrates over D_f fractal dimensions; 2π is absent because the relevant integration length is $\ln n$, not the circumference (A160).
- **Gauge sector** $U(1)$ and $SU(3)$. Was fully open. Now [B]: $U(1)_{\text{EM}}$ from flux quantisation $\Phi = n \cdot h/e$ on the torus; $SU(3)_C$ from linking number of three threads ($\dim(\mathfrak{su}(3)) = 8$ geometrically proved). $SU(2)_L$ from chiral projection (A095). Open edge: covariant derivatives D_μ not written out (A192).

- **Higgs-EFT 2.3% deviation.** Now explained [S]: SI values are scheme-dependent (1-loop) and fractal corrections in v not separable. $\xi_{\text{EFT}}/K_{\text{frak}}^{3/2} \approx 1.330 \times 10^{-4}$, residual only 0.3 % – but post-hoc correction, not proof (A190).
- **Gravitational quantisation.** Was a demand. Now declared: not required; $G = \xi^2/(4m_{\text{char}})$ is intrinsic (A140, A142).
- **ξ -proof demand.** Was open. Now declared: no proof possible or necessary; ξ is a fundamental parameter with five plausibility reasons (A015).

.3 The One Large Open Point

[STIPULATION] Above everything stands a residual that generates several others: the *measurement process* is not modelled (A090).

Which compact direction is unrolled and which becomes space is deferred to the measurement process and not resolved there. From this follow the openness of the unrolled direction (A060), the interpretational openness of Bell (A160), and the question of what distinguishes the observer as an embedded object (A090). Whoever closes this one point closes several.

.4 Residuals of the Foundation (A020–A090)

[STIPULATION] Absolute scale (A020, A085, A100, A130). The reduction to one dimension is derived; what size this dimension has is not. The same gap appears as the order of magnitude of ξ (A020), as the calibration of E_0 (A130), as the bridge constant v (A100) – one gap in several guises, not several gaps.

[STIPULATION] Factor 100 in K_{frak} (A040). Any integer factor closes the Pythagorean comma equally well; 100 is consistency, not necessity.

[STIPULATION] Non-closure quantity δ (A050). The recursion cannot fix it; free.

[S] $\mathbb{C}^3$ fibre and generations (A070). The assignment is interpretation, not derivation.

[STIPULATION] Holographic completeness type II (A090). Whether type-II loss is exactly reconstructible from boundary data is open.

[STIPULATION] Why P_+ and not P_- (A095). $g_R = 0$ is now [B]; why nature selects P_+ (orientation argument) remains [S].

.5 Residuals of the Sectors (A100–A192)

[STIPULATION] $\theta = 2/9$ (A120). Geometric source $C_3 \subset A_5$ known as a second witness; derivation from the stipulations is missing. Empirically sharp to seven places, structurally contingent.

[STIPULATION] Ladder exponents p_i (A100). Geometric sequence $p_{n+1} = \frac{2}{3}p_n$ established [K] for leptons and Up-type quarks (up to Charm); for Top ($p_t = -1/3$) and Down-type ($\Delta p = -1/2$) the geometric justification is missing.

[STIPULATION] $r = \sqrt{2}$ as stipulation (A110). Algebraic consequence: $Q = 2/3$. No derivation from ξ alone.

[STIPULATION] $\kappa = 4$ in $E_0 = e^2$ (A130). Whether the four follows from $\dim(T^4)$ or is an entry point remains open.

[STIPULATION] Characteristic mass in G (A145). $G = \xi^2/(4m_{\text{char}})$: m_{char} not uniquely fixed. Gravitational dynamics and inverse-square law completely missing (A140).

[STIPULATION] Quark sector (A150, A075). No counterpart to the circulant level; Hopf algebra and Feynman rules for interactions missing.

[STIPULATION] Neutrino sector (A150). Not testable.

[S] CHSH ξ -residual measured (A160, A165). IBM Heron r2 (May 2026) confirms Bell violation ($S = 2.74$); device-to-device variation $\sim 1.4\%$ exceeds ξ effect by factor 1400. Not resolvable in principle with NISQ hardware.

[S] Path assignment of entanglement (A160). Which closed paths correspond to which entanglement states: sketch.

[S] Landauer connection (A180). $\Delta w = 1 \leftrightarrow k_B T \ln 2$: connection justified [B], but that $\Delta w = 1$ is the *only* invariant unit has not been shown.

[STIPULATION] Gauge sector (A190, A192). $U(1)$ and $SU(3)$ from torus topology [B]; $SU(2)_L$ from chiral projection [B]. Open edge: covariant derivatives $D_\mu = \partial_\mu + ig A_\mu^a T^a$ not written out; coupling constants g_s, g_w, e not from ξ ; Higgs mechanism $SU(2) \times U(1) \rightarrow U(1)$ not from torus topology; CKM/PMNS missing.

[STIPULATION] Block-3 additions. Exponent $41/4$ in A265 calibrated to H_0 , not derived. $\alpha = 1$ stipulation (A267) worked out for EM only. Mass-free Dirac formulation (A263) for free fields only.

.6 Findings for the Correction Register

[STIPULATION] For booking in Dok. 190 (author's responsibility):

Symbol collision κ : Mass exponent $\kappa = 7$ (Dok. 009, correct) vs. energy-scale exponent (Dok. 101, erroneously 7 instead of 4). In Dok. 101, $\kappa = 7$ with $E_0 \approx 33.12$ gives $1/\alpha = 6.84$; correct is $\kappa = 4$, $E_0 = e^2$.

α labelling: "Derivation" (Dok. 011, 043) to be refined to "calibration / SI bookkeeping"; substance correct (A130).

.7 What the Residual Picture Shows

[B] Since the last edition the picture has shifted: the gauge sector is no longer fully open but at [B] level for $U(1)$ and $SU(3)$; chirality has been raised from [S] to [B]; the CHSH prefactor is geometrically justified. The two large construction sites remain – measurement process and complete gauge sector – but the gauge-sector site has shrunk.

The pattern is stable: a load-bearing core (geometry, units, lepton ratios, three gauge groups topologically), two large construction sites (measurement process, covariant derivatives and coupling constants), and a series of smaller declared stipulations.

.8 Open Questions

First, the collection is as complete as the sources. A230 bundles the residuals of the other documents; an open point not named there is also missing here.

Second, this document does not weight. Exception: the measurement process is explicitly the largest; all others stand without ranking (A200).

Third, "open" is not "unsolvable". The list says where the theory is incomplete, not whether the gaps are closable.

.9 Supersedes

This document collects the "Open Questions" sections of all A-documents (state: A-series extension 2026-07). It generates no new statements, but bundles existing ones and records points closed since the last edition.

.10 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.